

THE THEORY OF MEASURES AND INTEGRATION

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ABSTRACT. Lebesgue's integral patches up many of the defects of Riemann's integral, at the expense of the need to develop a notion of *measure*. These notes offer a broad and accelerated overview of the Lebesgue integral and measure theory with several exercises, open ended questions, and examples.

1. LECTURE 1: SOME MOTIVATION, BASIC DEFINITIONS AND RESULTS

1.1. Motivation. Length, Area, and Volume

We start with a simple question of *what is length?* We should be able to get some mathematical model for it. The standard objects of length measurements are one-dimensional, so it is fair to restrict ourselves to working in $\mathbb{R}$. In this setting, the objects of interest are, at their simplest, line segments, which are in turn naturally modeled by *intervals*.

So if we have an interval $[a, b] \subseteq \mathbb{R}$, it is then quite immediate to suggest its length:

$$\text{length of } [a, b] = b - a$$

1.2. Basic Definitions and Results. Our first definition formalizes the collections of sets that we can measure.

Definition 1.1. Given a set Ω , a σ -algebra of subsets of Ω is a family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ such that

- (i) $\emptyset, \Omega \in \mathcal{F}$,
- (ii) if $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, and
- (iii) if $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ then $\bigcup_{k \in \mathbb{N}} A_k \in \mathcal{F}$.

We'll say a pair $(\Omega, \mathcal{F})$ is a *measurable space*.

So σ -algebras are closed under countable unions, but they are also closed under countable intersections!

Question 1. Show that if $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ then

$$\bigcap_{k \in \mathbb{N}} A_k \in \mathcal{F}.$$

Let's pause here for a moment to ask, what kinds of σ -algebras are there?

Example 1.2. If Ω is a set, then $\mathcal{P}(\Omega)$ is a σ -algebra.

Actually *all* finite σ -algebras are built from these. (See Question 10)

But it would be dull if all σ -algebras were just power sets.

Definition 1.3. Given a set Ω and a family of subsets $\mathcal{S} \subseteq \mathcal{P}(\Omega)$, we define the σ -algebra generated by $\mathcal{S}$ to be the smallest σ -algebra containing $\mathcal{S}$. That is,

$$\sigma(\mathcal{S}) := \bigcap_{\substack{\mathcal{S} \subseteq \mathcal{F} \subseteq \mathcal{P}(\Omega) \\ \mathcal{F} \text{ is a } \sigma\text{-algebra}}} \mathcal{F}.$$

It is a simple exercise to show that these generated σ -algebras are indeed σ -algebras. But how are these useful? Well one example of when we deal with a set of subsets that is not inherently a σ -algebra, is the setting of a topological space.

Example 1.4. Let (X, τ) be a topological space. Then the σ -algebra generated by the topology (i.e. all the open sets), *Borel σ -algebra*, which we will denote by

$$\mathcal{B}(X) := \sigma(\tau).$$

For example, take $\mathbb{R}$ with its usual (metric) topology. Then the Borel σ -algebra is a non-trivial (i.e. not $\mathcal{P}(\mathbb{R})$) σ -algebra containing all open subsets of $\mathbb{R}$. We will revisit this non-trivialness after introducing the Lebesgue measure to show, using the Axiom of Choice to construct “Vitali sets”, which are non-Borel subsets of $\mathbb{R}$.

Question 2. Show that $\mathcal{B}(X)$ contains all closed sets, all F_σ sets, and all G_δ sets.

Curiosity 1. What is the cardinality of $\mathcal{B}(\mathbb{R})$? Seems like it contains a lot of sets, so it could be that $|\mathcal{B}(\mathbb{R})| = |\mathcal{P}(\mathbb{R})| = 2^{\mathfrak{c}}$.

However, it definitely does not contain all sets. So perhaps we have $|\mathcal{B}(\mathbb{R})| = \mathfrak{c}$?

If the latter is the case, that would be quite shocking, as all imaginable or constructable sets appear to be Borel, yet there is a whole size of infinity larger than the Borel sets composed of the unimaginable ones!

Question 3. Show that the Borel subsets of $\mathbb{R}$ (with its metric topology) are also generated by the following collections

- (i) $\{(a, b] : -\infty \leq a < b < \infty\}$,
- (ii) $\{(a, \infty) : a \in \mathbb{R}\}$
- (iii) $\{(a, b) : a < b \in \mathbb{R}\}$

In general, we have the following. The proof is a challenging exercise, but not necessary to get stuck on.

Fact 1. If $\mathcal{F}$ is an infinite σ algebra, then $\mathcal{F}$ is uncountable.

Definition 1.5. Given a measurable space $(\Omega, \mathcal{F})$, a (positive) *measure* on $\mathcal{F}$ (or on Ω if $\mathcal{F}$ is understood) is a function $\mu : \mathcal{F} \rightarrow [0, \infty]$ such that

- (i) $\mu(\emptyset) = 0$, and
- (ii) if $\{A_k\}_{k \in K} \subseteq \mathcal{F}$ with $A_i \cap A_j = \emptyset$ for all $i \neq j$, then

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \sum_{k=1}^{\infty} \mu(A_k).$$

A triple $(\Omega, \mathcal{F}, \mu)$ where $\mathcal{F}$ is a σ -algebra of subsets of Ω and μ is a measure on $\mathcal{F}$ will be called a *measure space*.

Our notions of length, area, and probability are all examples of measures, but their formal constructions will have to wait until we have some more tools. For now, let's see a couple of simple but elegant examples of measures that will end up bridging the gap between sums and integrals.

Example 1.6. Take $\Omega = \mathbb{N}$ and $\mathcal{F} = \mathcal{P}(\mathbb{N})$. Define the *counting measure* on $\mathbb{N}$, $\mu_{\text{count}} : \mathcal{P}(\mathbb{N}) \rightarrow [0, \infty]$ by

$$\mu_{\text{count}}(S) = |S|.$$

It is not hard to see that this satisfies all requirements of our definition.

In fact, we can define this same counting measure on any set, which we may encounter later. But the counting measure on $\mathbb{N}$ will be of particular importance to us.

One may reasonably call this a zero-dimensional volume measure, like length is to one-dimension and area is to two-dimensions!

Example 1.7. More fundamentally, for any Ω and any $a \in \Omega$, we have the *Dirac mass at a* measure, which is defined by

$$\delta_a(S) = \begin{cases} 1 & \text{if } a \in S \\ 0 & \text{otherwise} \end{cases}.$$

If measures are showing us how volume or mass is distributed, then this one says that all of it is concentrated at the point a . It is a point-mass!

Dirac masses are the simplest measures we can have but they are surprisingly useful.

Question 4. *The two measures we saw above are connected in a very direct way. Show that*

$$\mu_{\text{count}} = \sum_{n=1}^{\infty} \delta_n.$$

Question 5. *Motivated from the previous question, show that adding two measures results in a measure.*

Not all linear combinations of measures are permissible. Which ones are?

Such a space is often called a convex cone.

Question 6. *Show that every measurable space can be upgraded to a measure space with some measure.*

Curiosity 2. *Can we classify the set of measures somehow? Maybe in the simplest setting, say $|\Omega| < \infty$. Can we characterize all measures on $(\Omega, \mathcal{P}(\Omega))$? In this setting, if $\mathcal{F}$ is a sub- σ -algebra of $\mathcal{P}(\Omega)$ and μ is a measure on $(\Omega, \mathcal{F})$, can μ be extended to $\mathcal{P}(\Omega)$ for free?*

We also have measures associated to length and area, as we started out. But showing their existence is a bit more involved. For now, we delay this problem to a future section.

The requirement for the additivity of measure that we have disjoint sets is not a big limitation, for we can always *disjointify* a family, and at least get sub-additivity.

Question 7. *For a given family of sets $\{A_k\}_{k \in K}$, show that there is a disjoint family $\{B_k\}_{k \in K}$ such that*

$$\bigcup_{k=1}^n A_k = \bigcup_{k=1}^n B_k.$$

Question 8. *Show that, with the setup above,*

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \sum_{k=1}^{\infty} \mu(B_k).$$

Argue that for any countable collection $\{A_k\}_{k \in K} \subseteq \mathcal{F}$, we have sub-additivity of measure:

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) \leq \sum_{k=1}^{\infty} \mu(A_k).$$

We also quickly get using the same ideas the following computational tool for dealing with non-disjoint sets:

Theorem 1.8 (Inclusion-Exclusion). *Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $A, B \in \mathcal{F}$. Then*

$$\mu(A \cup B) = \mu(A) + \mu(B) - \mu(A \cap B).$$

Remark 1.9. One other objection one might have to the definition of measure is that we can seemingly define it equally fine for just finite unions. For example, one can define a *finitely-additive measure* on an algebra $\mathcal{F}$ of subsets of Ω as a map $\mu : \mathcal{F} \rightarrow [0, \infty]$ satisfying

- (i) $\mu(\emptyset) = 0$, and
- (ii) If $A_1, \dots, A_n \in \mathcal{F}$ are pairwise disjoint, then

$$\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n).$$

It is immediate that all measures in our original sense are finitely additive in this sense (why?). So, what benefit do we get from allowing for infinite unions?

A short possible justification is that by allowing countably infinities, we can deal with notions of limits and continuity, which vastly expands our scope. These kinds of arguments will be crucial for developing Lebesgue's integral later on.

The following is a result that is unique to countably-additive measures.

Theorem 1.10 (Continuity of Measure). *If $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ with $A_k \subseteq A_{k+1}$ for all $k \in \mathbb{N}$ then*

$$\mu \left(\bigcup_{k=1}^{\infty} A_k \right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Furthermore, if $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ with $A_k \supseteq A_{k+1}$ for all $k \in \mathbb{N}$, and if $\mu(A_1) < \infty$, then

$$\mu \left(\bigcap_{k=1}^{\infty} A_k \right) = \lim_{k \rightarrow \infty} \mu(A_k).$$

Question 9. *Show that without the assumption that $\mu(A_1) < \infty$ in the second part of the theorem above, the statement may not be true.*

1.3. Exercises.

Question 10. *We will show that all finite measurable spaces are of the form $(\Omega, \mathcal{P}(\Omega'))$. Let $(\Omega, \mathcal{F})$ be a measurable space where $|\mathcal{F}| < \infty$.*

- i *Say a nonempty subset $A \in \mathcal{F}$ is minimal if $B \in \mathcal{F}$ and $B \subseteq A$ implies that $B = \emptyset$ or $B = A$. Show that the collection of minimal elements of $\mathcal{F}$ partitions Ω (i.e. they are all disjoint and union to Ω).*
- ii *Find an appropriate set Ω' and a function $f : \Omega \rightarrow \Omega'$ such that $\hat{f} : \mathcal{F} \rightarrow \mathcal{P}(\Omega')$, defined by $\hat{f}(A) := f(A)$ is an order-isomorphism of $\mathcal{F}$ to $\mathcal{P}(\Omega')$.*

Question 11. *Prove Fact 1. (Hint: $|\mathcal{P}(\mathbb{N})| > |\mathbb{N}|$)*

Question 12. *Show that the σ -algebra generated by singletons on $\mathbb{R}$ is not $\mathcal{P}(\mathbb{R})$.*

Question 13. *Show that there are finitely-additive measures that are not countably-additive. (Hint: such a measure must fail the continuity theorem for some collection of sets)*

Question 14. *Show every measure on $\mathbb{N}$ is uniquely determined by a weight function $w : \mathbb{N} \rightarrow [0, \infty]$ in the sense that for any $S \subseteq \mathbb{N}$,*

$$\mu(S) = \sum_{n \in S} w(n).$$

Deduce which measures are σ -finite.

2. LECTURE 2: LEBESGUE'S INTEGRAL

Riemann's integral was a very successful approach that has some annoying limitations.

Motivation: balancing act

Question 15. *If we want to have our cake and eat it too, what functions satisfy having every preimage of an interval being also an interval?*

2.1. Permissible Functions. To allow for the chaos in Lebesgue's idea for the integral, we need to demand a certain property of our functions. In particular, for f to be integrable, a necessary condition is that the pre-images of intervals must have the ability to be measured. This condition is called *measurability*.

Definition 2.1. Given a measure space $(\Omega, \mathcal{F}, \mu)$, a function $f : \Omega \rightarrow \mathbb{R}$ is *measurable* if for all $B \in \mathcal{B}(\mathbb{R})$,

$$f^{-1}(B) \in \mathcal{F}.$$

For convenience, we denote the set of measurable functions on Ω by $\text{Meas}(\Omega, \mathbb{R})$. We will also sometimes restrict ourselves to positive measurable functions (that is, $f : \Omega \rightarrow [0, \infty)$ measurable), and we will denote this set by $\text{Meas}^+(\Omega, \mathbb{R})$.

Since we will be working with preimages, the following result will be of great use.

Question 16. *Let $f : X \rightarrow Y$. Show that*

- (i) $f^{-1}(Y) = X$, $f^{-1}(\emptyset) = \emptyset$,
- (i) if $S \subseteq Y$, then $f^{-1}(S^c) = f^{-1}(S)^c$, and
- (ii) if $\{S_n\}_{n=1}^{\infty} \subseteq Y$, then

$$f^{-1}\left(\bigcup_{n=1}^{\infty} S_n\right) = \bigcup_{n=1}^{\infty} f^{-1}(S_n).$$

Use the above observations to show that if $\mathcal{S}$ is a family of subsets of Y and $\sigma(\mathcal{S})$ is the σ -algebra generated by $\mathcal{S}$, then

$$\sigma(f^{-1}(\mathcal{S})) = f^{-1}(\sigma(\mathcal{S})),$$

where $\sigma(f^{-1}(\mathcal{S}))$ is the σ -algebra generated by $f^{-1}(\mathcal{S})$.

The above allows us to greatly reduce our efforts and pick suitably nice sets to show measurability on. The following is a nice example.

Question 17. *Show that all continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$ are Borel measurable. (Hint: It suffices to look at pre-images of open sets. Why?)*

Theorem 2.2. *If $(\Omega, \mathcal{F})$ is a measurable space, f, g measurable and $\lambda \in \mathbb{R}$ then we have*

- (i) λf is measurable
- (ii) $f + g$ is measurable
- (iii) fg is measurable
- (iv) $|f|$ is measurable
- (v) $\max\{f, g\}$ and $\min\{f, g\}$ are measurable.

Proof. We will show (ii) and leave the rest as an exercise for the reader which are a bit more direct.

We first show a naive approach which fails, in order to motivate a successful argument.

In order to show that $f + g$ is measurable, we need to show that for all Borel sets B , we have $(f + g)^{-1}(B) \in \mathcal{F}$. Writing out the set, we get

$$(f + g)^{-1}(B) = \{\omega \in \Omega \mid f(\omega) + g(\omega) \in B\}.$$

However, even in the case when B is a singleton, say $\{a\}$, we have $f(\omega) + g(\omega) = a$, which tells us nothing about $f(\omega)$ or $g(\omega)$. In principle, either one could be any real number, as long as the other

compensates. With this observation, we may write (check this)

$$(f + g)^{-1}(B) = \bigcup_{r \in \mathbb{R}} f^{-1}(\{r\}) \cap g^{-1}(B - r).$$

Here $B - r = \{x \in \mathbb{R} | x = b - r \text{ for some } b \in B\}$ is a Borel set. (why?) However, if you are comfortable with σ -algebras, you might recognize that the above attempt doesn't quite work, as it is an uncountable union! If only we could make it countable somehow.

Luckily, we can make use of the properties of the Borel σ -algebra, in particular that they are generated by many families of *nice* sets (see Question 3). We argue that since the intervals (a, ∞) generate $\mathcal{B}(\mathbb{R})$, then it suffices to show that $(f + g)^{-1}((a, \infty)) \in \mathcal{F}$ for all $a \in \mathbb{R}$ to guarantee measurability. (why?)

With this strategy in mind, we can reduce our uncountable union down to a countable one with the following clever idea (check this equality):

$$(f + g)^{-1}((a, \infty)) = \bigcup_{k \in \mathbb{Z}} f^{-1}((-\infty, k]) \cap g^{-1}((a - k, \infty)).$$

By measurability of f and g and the closure under countable unions and intersections, we are thus guaranteed the measurability of $f + g$. $\square$

Question 18. Find a function f such that $|f|$ is measurable, but f is not.

Remark 2.3. Properties (i), (ii), and (iii) in the theorem above say that $\text{Meas}(\Omega, \mathbb{R})$ is an algebra. Property (v) guarantees us that it is also a lattice. So $\text{Meas}(\Omega, \mathbb{R})$ has a decently rich algebraic structure to it.

In addition, the question preceding the theorem above shows us that when $(\Omega, \mathcal{F}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then

$$C(\mathbb{R}, \mathbb{R}) \subseteq \text{Meas}(\mathbb{R}, \mathbb{R}).$$

This containment is strict. There is a much richer collection of measurable functions than continuous ones.

Example 2.4. A simple example of discontinuous measurable functions are the *characteristic functions*. Let $(\Omega, \mathcal{F})$ be a measurable space, and let $E \in \mathcal{F}$. Then define the associated characteristic function, χ_E as follows.

$$\chi_E(\omega) = \begin{cases} 1 & \text{if } \omega \in E \\ 0 & \text{otherwise} \end{cases}.$$

It is easy to see that such a function is measurable. Indeed, let B be any Borel subset of $\mathbb{R}$, then we have

$$\chi_E^{-1}(B) = \begin{cases} \Omega & \text{if } 0, 1 \in B \\ E & \text{if } 1 \in B \text{ and } 0 \notin B \\ E^c & \text{if } 1 \notin B \text{ and } 0 \in B \\ \emptyset & \text{otherwise} \end{cases}.$$

So for example, if $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, and $E = \mathbb{Q}$, then $\chi_{\mathbb{Q}}$, the characteristic function of the rational numbers, also known as the Dirichlet function, is everywhere discontinuous but measurable.

Curiosity 3. There are many questions we can ask about the nature of measurable functions to understand them better. For starters, what is $|\text{Meas}(\Omega, \mathbb{R})|$?

Also, how different are measurable functions from continuous functions, really? Though the Dirichlet function is discontinuous everywhere, it seems that if we changed its function values on a set of zero measure (the rationals), we could get a continuous function.

Does $\text{Meas}(\mathbb{R}, \mathbb{R})$ have any kind of natural metric or topology associated to it, to tell us when measurable functions are close? (We will soon see that in fact this space is closed under point-wise limits!)

Let's get on with defining our integrals. We proceed in stages, starting with the simplest functions, and using these to build up to the most general cases. For the following development, fix a measure space $(\Omega, \mathcal{F}, \mu)$.

Step One: Characteristic Functions

Let $E \in \mathcal{F}$ and consider its characteristic function χ_E . If we apply Lebesgue's approach to defining the integral, we come up with the following reasonable definition for the integral of E .

$$\int \chi_E d\mu := \mu(E).$$

So far so good, but characteristic functions are a very limited class of functions. We can improve this immediately by extending this definition linearly, as follows.

Step Two: Simple Functions

Definition 2.5. A measurable function $\varphi : \Omega \rightarrow \mathbb{R}$ is called *simple* if $|\text{Ran } \varphi| < \infty$.

Question 19. Show that simple functions are exactly the linear combinations of characteristic functions.

Let φ be simple with $\text{Ran } \varphi = \{r_1, \dots, r_n\}$. Then define

$$\int \varphi d\mu := \sum_{k=1}^n r_k \cdot \mu(\varphi^{-1}(\{r_k\})).$$

This is still giving us a seemingly very small class of functions to work with, but now that we have these simple functions, we can use these to approximate more arbitrary functions. We start with the positive ones.

Step Three: Positive Functions

Even with Riemann's approach, we approximated the area under a curve with simple functions which were constant on subintervals. Lebesgue's idea of chopping up the range allows us to use simple functions which are constant on measurable partitions, expanding the class of approximation functions significantly. But the following definition closely parallels the Riemann approach.

Definition 2.6. If $f : \Omega \rightarrow [0, \infty)$ is measurable define

$$\int f d\mu = \sup_{\substack{\varphi \text{ simple} \\ \varphi \leq f}} \int \varphi d\mu.$$

Question 20. Could we have defined this as an infimum instead, over all simple functions larger than f ?

Step Four: Generalizing

To extend this integral to a more general class of measurable functions, we use the ability to write a function in terms of positive ones. The following lemma takes care of this.

Lemma 2.7. If $f \in \text{Meas}(\Omega, \mathbb{R})$ then there exist $f_+, f_- \in \text{Meas}^+(\Omega, \mathbb{R})$ such that

$$f = f_+ - f_-$$

Proof.

□

Now one might naturally come up with the following extension of the integral to f :

$$\int f d\mu := \int f_+ d\mu - \int f_- d\mu.$$

This is indeed how we do this, however we offer one caveat. Namely, the integrals may not have been finite, so there is potential in the above that we are left with the paradoxical

$$\int f d\mu = \infty - \infty.$$

To avoid this, we simply stipulate that for the integral of f to be well defined, then at most one of $\int f_+ d\mu$ and $\int f_- d\mu$ is infinite.

Definition 2.8. We will say a measurable function f is *integrable* if

$$\int |f| d\mu = \int f_+ d\mu + \int f_- d\mu < \infty.$$

Denote by $\mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ the space of integrable functions on $(\Omega, \mathcal{F}, \mu)$.

Now that the integral is properly defined, let's take a moment to verify some basic properties. The following lemma says that the integral is both *monotone* and *linear*.

Lemma 2.9. If f, g are integrable with $f \leq g$, then

$$\int f d\mu \leq \int g d\mu.$$

Furthermore, if $\lambda \in \mathbb{R}$, we have

$$\int (\lambda f + g) d\mu = \lambda \int f d\mu + \int g d\mu.$$

The Lebesgue integral respects limits very nicely too. We start with measurability.

Theorem 2.10. If $\{f_n\}_{n \in \mathbb{N}}$ is a sequence of measurable functions which converges point-wise to f , then f is measurable.

Theorem 2.11 (Monotone Convergence Theorem). If $\{f_n\}_{n \in \mathbb{N}}$ is a family of non-negative, measurable functions such that $f_{n+1} \geq f_n$ and $f(x) := \sup_{n \in \mathbb{N}} f_n(x)$, then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu.$$

This discussion has been largely abstract. Let's see if we can compute some examples!

Example 2.12. Let Ω be any set and let $\mathcal{F} = \mathcal{P}(\Omega)$. Fix $a \in \Omega$ and consider the Dirac mass δ_a . Then integration against δ_a has a surprising form: for any $f \in \text{Meas}(\mathbb{R}, \mathbb{R})$, we have

$$\int f d\delta_a = f(a).$$

So integration of a function against the Dirac mass at a is just evaluation of f at a !

This is not too difficult to show directly from the definition, see if you can.

Example 2.13. Let $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, and take the counting measure μ_{count} . Then, for $f : \mathbb{N} \rightarrow \mathbb{R}$ integrable with respect to μ_{count} , we have

$$\int f d\mu_{\text{count}} = \sum_{n=1}^{\infty} f(n).$$

Furthermore, f is integrable with respect to μ_{count} if and only if its corresponding series is absolutely summable.

So every time you evaluated a function or took a summation, you were doing integrals!

2.2. Exercises.

Question 21. Show that if $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then it is Lebesgue integrable, and furthermore its Lebesgue integral equals its Riemann integral.

See if you can generalize this to all Riemann integrable functions on $[a, b]$.

Question 22. Show that $\mathcal{L}^1(\Omega, \mathcal{F}, \mu)$ is a vector space equipped with a semi-norm

$$\|f\|_1 = \int |f| d\mu.$$

Why is it not a norm in general?

When taking a quotient of $\mathcal{L}^1$ by the relation $f \sim g$ if $\|f - g\|_1 = 0$, we get the quotient space

$$L^1(\Omega, \mathcal{F}, \mu) = \mathcal{L}^1(\Omega, \mathcal{F}, \mu) / \sim.$$

It is a nice fact that this turns out to be a Banach space.

3. LECTURE 3: THE RIESZ-MARKOV-KAKUTANI REPRESENTATION THEOREM

Coming soon!

4. LECTURE 4: RADON-NIKODYM DERIVATIVES

Given a measure space $(\Omega, \mathcal{F}, \mu)$ and a measurable function $f : \Omega \rightarrow \mathbb{R}$, we can *push forward* μ to a measure on the real line. For each Borel set B , define

$$\mu_f(B) := \mu(f^{-1}(B)).$$

It is straightforward to verify that this is a Borel measure on $\mathbb{R}$ due to the measurability of f .

In probability theory (see Supplement 1 below), we start with a *probability space* $(\Omega, \mathcal{F}, P)$, where P is just a measure with the added constraint that $P(\Omega) = 1$. Measurable functions are then called *random variables*, and the pushforward measure described above is called the *distribution* of the random variable f .

Question 23. *Do every two different measurable functions yield different pushforward measures on $\mathbb{R}$?*

The pushforward allows us to turn an integral over our sample space Ω into an integral over $\mathbb{R}$.

Question 24. *Show that*

$$\int_{\Omega} f(\omega) d\mu(\omega) = \int_{\mathbb{R}} x d\mu_f(x).$$

In probability theory, f is called a random variable and then the quantity above is the expected value of f .

As you may have noticed, this transformation wasn't too profound. However, we also often perform another little trick. Suppose for example that f has a *standard normal distribution*. Then

$$\int_{\Omega} f(\omega) dP(\omega) = \int_{\mathbb{R}} x d\mu_f(x) = \int_{\mathbb{R}} x \left(\frac{e^{-\frac{x^2}{2}}}{\sqrt{2\pi}} \right) dx.$$

Here dx refers to the Lebesgue measure on $\mathbb{R}$. This change in the measure of integration is useful enough to appear in most expected value calculations.

It would be nice if we could always just convert back to a nice measure like the Lebesgue measure to perform our calculations and analysis. However, can we really do this all the time?

Our hope dies very quickly when considering a simple measure like a Dirac mass. For example we saw in Lecture 2 that for any measurable f ,

$$\int_{\mathbb{R}} f(x) d\delta_a = f(a).$$

So in particular,

$$\int_{\mathbb{R}} \chi_{\{a\}} d\delta_a = 1.$$

But if we try to change this to an integral against the Lebesgue measure, we run into problems. Note that $\chi_{\{a\}} f(x) = f(a) \chi_{\{a\}}$, so that

$$\int_{\mathbb{R}} \chi_{\{a\}} f(x) dx = f(a) \int_{\mathbb{R}} \chi_{\{a\}} dx = f(a) \lambda(\{a\}) = 0.$$

We thus arrive at a contradiction if we suppose that we can find some $f(x)$ such that

$$\int_{\mathbb{R}} g(x) d\delta_a = \int_{\mathbb{R}} g(x) f(x) dx$$

for all integrable $g(x)$.

What went wrong in the above counter-example was the existence of a set $(\{a\})$ which was massive for δ_a , but massless for λ . The existence of such sets yields contradictions in a very similar way to what is seen above.

In general, if μ_1 and μ_2 are such that we could find an f with

$$\int_{\Omega} g d\mu_1 = \int_{\Omega} g f d\mu_2,$$

for all g , then by choosing $g = \chi_E$, we see that it must be the case that whenever $\mu_2(E) = 0$, it follows that $\mu_1(E) = 0$ (check this).

This leads us to the following definition.

Definition 4.1. Given two measures μ_1 and μ_2 on $(\Omega, \mathcal{F})$, we say that μ_1 is *absolutely continuous with respect to* μ_2 , and write $\mu_1 \ll \mu_2$ if whenever $S \in \mathcal{F}$ with $\mu_2(S) = 0$, it follows that $\mu_1(S) = 0$.

Example 4.2. Motivated by considerations above, it is easy to see that for any $a \in \mathbb{R}$,

$$\lambda \ll \delta_a \quad \text{but} \quad \delta_a \not\ll \lambda.$$

Curiosity 4. One might ask the question, is there a measure, μ , on Ω such that for every other measure ν , we have $\nu \ll \mu$? What about $\mu \ll \nu$?

For the first, we note that if $\mu \gg \nu$ for every measure ν , then, for example $\mu \gg \delta_a$ for every $a \in \Omega$. So in particular, we must have that $\mu(\{a\}) > 0$ for every $a \in \Omega$. One such measure for which this holds is the counting measure! Can you show that every other measure on Ω is absolutely continuous with respect to the counting measure?

For the second, we may note simply that the zero measure is zero on every subset. So if $\mu \ll \nu$ for every measure, then $\mu \ll 0$, which forces us to conclude that μ is in fact the zero measure. It is immediate that the zero measure also satisfies the property of being absolutely continuous with respect to every measure.

Remark 4.3. We note that while absolute continuity is transitive, it is not anti-symmetric. That is, we can have two different measures μ_1 and μ_2 such that $\mu_1 \ll \mu_2$ and $\mu_2 \ll \mu_1$. One simple example of this is to take any measure μ_1 and let $\mu_2 = c\mu_1$ for any $c > 0$.

In this case, since we can select $c > 1$ or $c < 1$, we see that the relation $\ll$ is not so much about one measure being larger than another in a “point-wise” sense. Rather, it’s more of a comparison of the relative *supports* of the two measures (i.e. on which sets the mass lives).

What is remarkable, is that if our measures are relatively well behaved (σ -finite), this condition of absolute continuity is all we need! The following is the main result of this lecture.

Theorem 4.4 (Radon-Nikodym Theorem). *If μ_1 and μ_2 are σ -finite measures on $(\Omega, \mathcal{F})$ such that $\mu_1 \ll \mu_2$ then there exists a measurable $f : \Omega \rightarrow [0, \infty)$ such that for all $E \in \mathcal{F}$,*

$$\mu_1(E) = \int_E f \, d\mu_2.$$

Furthermore, if $g : \Omega \rightarrow [0, \infty)$ is measurable such that $\mu_1(E) = \int_E g \, d\mu_2$, then $g = f$ almost everywhere.

See Question 26 below for an example of how the Radon-Nikodym theorem fails when we aren’t dealing with σ -finite measures.

Question 25. *This might not look like what is promised, however, show that the above implies that*

$$\int_{\Omega} g d\mu_1 = \int_{\Omega} g f d\mu_2.$$

(Hint: Build up to f through characteristic functions, then simple functions, then limits as in the development of the integral.)

4.1. Exercises.

Question 26. We saw that every measure is absolutely continuous with respect to the counting measure μ_{count} . Use this fact about the Lebesgue measure to conclude that the Radon-Nikodym theorem is not necessarily true when our measures are not σ -finite.

Question 27. Recall from Question 14 (Lecture 1), that every σ -finite measure μ on $\mathbb{N}$ is determined by a weight function $w : \mathbb{N} \rightarrow [0, \infty)$. Furthermore, we saw that every measure is absolutely continuous with respect to the counting measure, which on $\mathbb{N}$, is σ -finite itself. Show that the Radon-Nikodym derivative of μ with respect to the counting measure is exactly the weight function w .

5. LECTURE 5: SIGNED MEASURES

One idea that may have sprung to mind during the discussion of the Radon-Nikodym derivatives of the last lecture, was “what if the derivative function f was an arbitrary integrable function?”. In particular, what happens if we define

$$\nu(E) = \int_E f \, d\mu$$

where now $f : \Omega \rightarrow \mathbb{R}$ is simply integrable, not necessarily non-negative?

On one hand we might easily dismiss this, as, for example, we could let $E = f^{-1}((-\infty, 0))$, and then we would have to conclude that if $\mu(E) > 0$, then

$$\nu(E) < 0.$$

However, this object ν still behaves like a measure in the sense that $\nu(\emptyset) = 0$, and ν is countably additive.

Question 28. *Show that ν , defined as above, is countably additive. You will need a limit theorem for integrals for this.*

So ν is a measure, that measures some sets as negative. We call such an object a *signed measure*. More formally,

Definition 5.1. A set function $\mu : \mathcal{F} \rightarrow [-\infty, \infty)$ or $\mu : \mathcal{F} \rightarrow (-\infty, \infty]$ a *signed measure* if

- (1) $\mu(\emptyset) = 0$, and
- (2) whenever $\{A_k\}_{k \in \mathbb{N}} \subseteq \mathcal{F}$ are pairwise disjoint, then

$$\mu\left(\bigcup_{k=1}^{\infty} A_k\right) = \sum_{k=1}^{\infty} \mu(A_k).$$

Remark 5.2. In the definition, we disallow μ to take on both positive and negative infinite values. For if E and F are pairwise disjoint such that $\mu(E) = \infty$ and $\mu(F) = -\infty$, then it is not clear how we compute

$$\mu(E \cup F) = \mu(E) + \mu(F) = \infty - \infty = ?$$

Signed measures might seem like generalization for the sake of generalization, however they seem to naturally sit in some gaps left behind by our discussions of positive measures. For example,

Theorem 5.3 (Hahn Decomposition Theorem). *For any signed measure μ , there exists a positive set P and negative set N such that $P \cup N = \Omega$.*

Theorem 5.4 (Jordan Decomposition Theorem).

- Complex measures

6. LECTURE 6: PRODUCT MEASURES

7. SUPPLEMENT 1: PROBABILITY THEORY

Measure theory is the formal basis of probability theory where we only make one additional assumption.

Definition 7.1. A measure space $(\Omega, \mathcal{F}, \mu)$ is called a *probability space* if $\mu(\Omega) = 1$.

Note by $\text{Prob}(\Omega, \mathcal{F})$ the set of probability measures on $\mathcal{F}$.

Remark 7.2. This additional assumption on the measure for probability spaces means that naively taking positive linear combinations of probability measures need not yield a probability measure. However, some such combinations still do. In particular, the *convex* combinations.

Question 29. Show that convex combinations of probability measures are probability measures. That is, $\text{Prob}(\Omega, \mathcal{F})$ is a convex subset of $\mathcal{M}(\Omega, \mathcal{F})$.

Curiosity 5. So the set of probability measures on Ω is guaranteed to be convex. One natural place to look next is for extreme points. These are the points that cannot be written as a non-trivial convex combination of two other distinct points. For example, a rectangle, including its inner area is convex with its vertices as its extreme points. This leads us to the question: what are the extreme points of $\text{Prob}(\Omega, \mathcal{F})$? Maybe the case when Ω is finite is more tractable?

In probability theory, we are usually concerned with studying random variables. However, we have also already seen these, as random variables are simple measurable functions!

Definition 7.3. Given a probability space $(\Omega, \mathcal{F}, P)$, a *random variable* is a (Borel) measurable function $f : \Omega \rightarrow \mathbb{R}$.

The “randomness” in a random variable comes from ignoring their domain and instead looking at the “distribution” of their outputs. Even though a function is by definition deterministic, if we blind ourselves to the inputs of the function, then the outputs will appear “random” (but following a distribution).

Definition 7.4. Given a random variable $X : \Omega \rightarrow \mathbb{R}$, the *distribution of X* is the (probability) measure $\mu_X : \mathcal{B}(\mathbb{R}) \rightarrow [0, 1]$ defined by

$$\mu_X(B) := P(X^{-1}(B)), \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

Remark 7.5. So in particular, distributions of random variables are actually just (Borel) probability measures on $\mathbb{R}$! It is not unfair to say that studying probability theory is studying the space $\text{Prob}(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. In fact, this space has certain algebraic structure to it, in the form of *convolution* (the distribution you get when you add two independent random variables with corresponding distributions). We defer a closer look into this until later.

Associated to distributions are the *distribution functions*. However, these are simply properties of measures on $\mathbb{R}$.

Definition 7.6. For $\mu \in \mathcal{M}(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, define its *cumulative distribution function*, $F_\mu : \mathbb{R} \rightarrow [0, \infty]$ by

$$F_\mu(x) := \mu((-\infty, x]).$$

Fact 2. The density function associated to μ is the Radon-Nikodym derivative of the measure μ with respect to the Lebesgue measure. (See Lecture 4.) As a consequence, this does not exist when μ has atoms (i.e. points $x \in \mathbb{R}$ such that $\mu(\{x\}) > 0$), but often does exist even when the cumulative distribution function F_μ is not differentiable!

Of course, when F_μ is differentiable, then the density function f_μ is simply the derivative of F_μ .

The next thing one picks up in an introductory probability course is the notion of expected value for a random variable. If not introduced via measure theory, one might have to split our definition up into a case for discrete random variables and a case for continuous random variables. However, with the Lebesgue integral, we no longer need to make this distinction.

Definition 7.7. Given a probability space $(\Omega, \mathcal{F}, P)$ and a random variable $X : \Omega \rightarrow \mathbb{R}$, its *expected value* is the quantity

$$\mathbb{E}(X) := \int_{\Omega} X dP = \int_{\mathbb{R}} x d\mu_X.$$

Question 30. *Verify that the equality in the definition above is true.*

Question 31. *Show that if $\text{Ran } X = \{1, 2, 3\}$ and*

$$P(X^{-1}(\{1\})) = p_1, \quad P(X^{-1}(\{2\})) = p_2, \quad \text{and} \quad P(X^{-1}(\{3\})) = p_3,$$

then

$$\mathbb{E}(X) = 1 \cdot p_1 + 2 \cdot p_2 + 3 \cdot p_3.$$